

TAM THUC STRATEGEM

Backend Technical Specification v1.0

1. TỔNG QUAN KIẾN TRÚC BACKEND

Backend được thiết kế theo kiến trúc microservices-like (hoặc modular monolith ở MVP) với các module chính:

- Calculation Engines (3 hệ thống độc lập nhưng có interface chung)
- Rule Engine & Pattern Matching
- RAG Knowledge Service
- User & Query Management
- Report Generation Service

2. TECH STACK CHI TIẾT

- Framework: FastAPI (Python 3.11+)
- Database: PostgreSQL 15+ (users, queries, history, patterns)
- Vector DB: Chroma or Pinecone (knowledge embeddings)
- Cache: Redis
- Task Queue: Celery + Redis (cho long-running tasks như batch reports)
- Calendar: Custom hoặc adapt từ lunar calendar libs + astronomical calculations cho Jieqi
- Auth: JWT + OAuth2 (Google/Apple cho social login)

3. DATABASE SCHEMA (CHÍNH)

Users: id, email, password_hash, birth_data (encrypted JSON), preferences, created_at
Queries: id, user_id, query_type, input_data (JSON), systems_used, result_summary, created_at
Charts: id, query_id, system (qimen/liuren/taiyi), chart_data (JSON full), patterns_detected (JSON)
Knowledge_Patterns: id, system, pattern_name, conditions (JSON), meaning_classical, meaning_modern, citations, confidence
Reports: id, query_id, report_type, content (Markdown or JSON), pdf_path, generated_at

4. API ENDPOINTS (CHÍNH - REST + một số WebSocket)

Public/Auth:
POST /auth/register, /auth/login, /auth/refresh

Core Calculation:
POST /api/v1/calculate/qimen
POST /api/v1/calculate/liuren
POST /api/v1/calculate/taiyi
POST /api/v1/calculate/all (cross validation)

Strategic Tools:
POST /api/v1/timing/optimize
POST /api/v1/scenario/compare

Knowledge & Learning:
GET /api/v1/knowledge/patterns?system=qimen
GET /api/v1/learning/simulator/steps?qimen

Reports:
POST /api/v1/reports/generate
GET /api/v1/reports/{id}/download

5. CALCULATION ENGINE MODULE

Mỗi hệ thống có module riêng (qimen_engine.py, liuren_engine.py, taiyi_engine.py) với interface chung:
`def calculate(input_data: dict) -> ChartResult`

Sử dụng/adapt open-source đã validate (qimen, kintaiyi, kinliuren) + custom code cho edge cases và metadata.

Tất cả output được normalize thành common Chart schema để dễ cross-validation.

6. RULE ENGINE & PATTERN MATCHING

Patterns được lưu trong DB (Knowledge_Patterns) dưới dạng JSON conditions.

Rule Engine quét chart_data → match conditions → trả về list patterns + scores.

Hỗ trợ versioning patterns để dễ update theo expert feedback.

7. RAG KNOWLEDGE SERVICE

- Embeddings: OpenAI text-embedding-3-small hoặc local model (nếu cần on-prem).
- Retrieval: Top-k relevant chunks (classical excerpts + pattern explanations).
- Prompt construction: System prompt + retrieved context + user query + persona level.
- Output parsing: Structured JSON (interpretation, citations, confidence, suggestions).

8. SECURITY & COMPLIANCE

- Personal data (birth_data) encrypted at rest (AES-256).
- All API over HTTPS.
- Rate limiting + abuse detection.
- Audit logs cho sensitive actions.
- GDPR/VN PDPA compliance (right to be forgotten, data export).

9. DEPLOYMENT & SCALING

- Docker + Kubernetes.
- Horizontal scaling cho calculation services (stateless).
- Celery workers cho background tasks.
- Monitoring: Prometheus + Grafana + Sentry.
- CI/CD: GitHub Actions → staging → production.

10. TESTING STRATEGY (BACKEND)

- Unit tests: Mỗi engine function (pytest, >80% coverage).
- Integration tests: Full flow calculate → patterns → RAG.
- Validation tests: So sánh output với 100+ classical examples (dataset curated).
- Load tests: K6 hoặc Locust cho 1000 concurrent chart generations.
- Expert review: Master review output của 50 random queries mỗi release.

Tài liệu này sẽ được cập nhật theo từng phase phát triển. Version control qua Git + changelog.